

Introduction to Machine Learning

Machine Learning

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Learning from Data

- Can we learn about the world around us using data?
- Model building from data
 - Take data as input
 - Find patterns in the data
 - Summarize the pattern in a mathematically precise way
- Machine learning automates this model building.

The Challenge

- Data unfortunately contains noise. If not, machine learning would be trivial!
- Think of $\text{Data} = \text{Information} + \text{Noise}$
- The challenge is to identify the information content and distill away the noise.
- To help do this, machine learning uses a train and test approach.

Over fitting Vs under fitting

- If the model we finish with ends up
 - modeling the noise as well, we call it “over fitting” - bad for prediction!
 - not modeling all the information, we call it “under fitting” - bad for prediction!
- The hope is that the model that does the best on testing data manages to capture/model all the information but leave out all the noise.

Machine Learning tasks

1. Supervised learning: Building a mathematical model using data that contains both the inputs and the desired outputs (ground truth).

- Examples:
 - Determining if an image has a horse. The data would include images with and without the horse (the input), and for each image we would have a label (the output) indicating if there is a horse in that image.
 - Determining if a client might default on a loan
 - Determining if a call center employee is likely to quit
- Since we have desired outputs, model performance can be evaluated by comparisons.

Machine Learning Tasks

2. **Unsupervised learning:** Building a mathematical model using data that contains only inputs and no desired outputs.

- Used to find structure in the data, like grouping or clustering of data points. To discover patterns and group the inputs into categories.
- Example: an advertising platform segments the population into smaller groups with similar demographics and purchasing habits. Helping advertisers reach their target market with relevant ads.
- Since no labels are provided, there is no specific way to compare model performance in most unsupervised learning methods.

Tools and techniques

- Supervised learning
 - Regression: desired output is a continuous number
 - Classification: desired output is a category
- Unsupervised learning
 - Clustering: Grouping data
 - Dimensionality reduction: Compressing data
 - Association rule learning: If X then Y